

DAUPHIN, MB, Canada

WMO: 715500

Lat: 51.1003N

Lon: 100.0569W

Elev: 305

StdP: 97.72

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.6	-29.1	-36.4	0.1	-31.4	-33.8	0.2	-28.8	13.1	-11.1	12.0	-9.7	3.6	250	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	29.9	20.8	28.2	19.5	26.6	18.6	22.4	27.7	21.1	26.3	19.9	24.8	5.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.8	25.5	19.2	14.5	24.0	17.9	13.4	22.9	67.5	27.6	62.7	26.4	58.4	24.9	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.3	9.2	DB	-35.7	33.6	2.4	1.9	-37.5	35.0	-38.9	36.1	-40.2	37.1	-42.0	38.5	(4)
(5)			WB	-35.8	25.0	2.4	1.5	-37.6	26.1	-39.0	26.9	-40.3	27.8	-42.1	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-15.5	-13.5	-6.5	3.1	10.3	16.3	19.1	18.0	12.9	5.0	-4.5	-12.5	(6)
(7)		DBStd	13.76	8.48	7.75	7.97	6.37	5.19	3.92	3.20	3.49	4.55	4.99	6.94	8.05	(7)
(8)		HDD10.0	3567	789	659	512	218	60	2	0	0	22	170	435	699	(8)
(9)		HDD18.3	5809	1048	893	769	457	253	82	30	49	171	415	684	957	(9)
(10)		CDD10.0	925	0	0	0	12	70	193	281	247	108	14	1	0	(10)
(11)		CDD18.3	125	0	0	0	0	4	22	53	38	8	0	0	0	(11)
(12)		CDH23.3	1386	0	0	0	6	82	238	506	431	117	6	0	0	(12)
(13)		CDH26.7	343	0	0	0	1	17	56	122	121	26	1	0	0	(13)
(14)	Wind	WSAvg	4.2	4.3	4.3	4.1	4.4	4.6	4.1	3.7	3.7	4.1	4.3	4.3	4.2	(14)
(15)	Precipitation	PrecAvg	513	21	17	28	34	56	79	70	67	64	33	24	22	(15)
(16)		PrecMax	727	52	43	77	90	132	200	198	180	172	99	62	45	(16)
(17)		PrecMin	303	5	4	2	0	6	7	27	5	8	3	0	8	(17)
(18)		PrecStd	96	12	10	17	24	32	48	38	46	41	27	14	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	4.9	13.0	23.8	28.9	31.1	31.7	32.7	29.7	23.0	14.7	6.9	(19)
(20)			MCWB	2.2	2.2	6.9	12.9	16.2	21.1	23.5	21.5	18.8	14.2	8.6	3.3	(20)
(21)		2%	DB	2.5	2.4	9.2	19.7	25.6	28.2	29.6	29.7	26.7	18.8	10.3	3.6	(21)
(22)			MCWB	0.7	0.7	5.2	10.7	14.3	18.6	21.6	20.4	17.5	11.1	5.9	1.1	(22)
(23)		5%	DB	-0.1	0.6	6.1	16.7	22.8	26.0	27.9	27.5	24.1	15.8	7.0	1.0	(23)
(24)			MCWB	-1.4	-0.7	3.3	9.3	13.2	17.8	20.4	19.6	16.1	10.0	3.7	-0.9	(24)
(25)		10%	DB	-2.9	-1.8	3.7	13.5	20.2	24.1	26.2	25.6	21.3	13.0	4.5	-1.7	(25)
(26)			MCWB	-4.0	-2.9	1.8	7.6	12.1	16.8	19.6	18.5	14.8	8.5	2.3	-2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.5	8.1	13.6	18.0	23.2	24.7	23.6	20.7	14.9	9.0	3.1	(27)
(28)			MCDWB	4.3	4.1	11.9	21.7	25.3	28.6	29.7	29.4	26.0	21.3	13.8	6.6	(28)
(29)		2%	WB	0.8	0.8	5.4	11.2	15.8	20.5	22.8	21.7	18.6	12.3	6.3	1.3	(29)
(30)			MCDWB	2.4	2.2	8.8	18.5	22.8	25.9	27.9	27.6	24.5	17.2	9.8	3.6	(30)
(31)		5%	WB	-1.4	-0.7	3.5	9.6	14.3	18.9	21.5	20.6	17.1	10.6	4.0	-0.8	(31)
(32)			MCDWB	0.0	0.6	6.0	15.8	21.0	23.8	26.4	26.0	22.6	14.6	6.3	0.9	(32)
(33)		10%	WB	-4.0	-2.9	1.9	7.9	12.8	17.7	20.3	19.3	15.4	9.0	2.3	-2.6	(33)
(34)			MCDWB	-3.0	-1.8	3.6	13.4	18.8	22.7	25.0	24.4	20.5	12.8	4.4	-1.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	10.1	11.7	12.7	11.2	11.7	12.5	12.1	9.8	8.3	8.6	(35)
(36)			MCDBR	10.2	9.8	11.2	16.9	16.8	14.0	13.8	15.5	16.7	15.1	11.9	10.4	(36)
(37)			MCWBR	8.9	8.3	8.2	9.7	8.6	7.1	7.5	8.1	8.9	9.1	8.3	8.6	(37)
(38)		5% WB	MCDBR	10.0	9.4	10.7	15.6	14.4	11.9	12.7	13.8	15.0	13.6	11.2	10.1	(38)
(39)			MCWBR	8.8	8.2	8.0	9.3	8.2	6.9	7.7	8.0	8.8	8.5	8.2	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.263	0.267	0.305	0.337	0.350	0.366	0.376	0.372	0.340	0.293	0.263	0.243	(40)	
(41)		taud	2.253	2.282	2.297	2.395	2.417	2.400	2.394	2.398	2.473	2.566	2.467	2.343	(41)	
(42)		Ebn at Noon	761	872	895	905	905	889	873	858	852	844	781	744	(42)	
(43)		Edh at Noon	67	86	104	107	109	112	111	106	88	66	55	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.27	2.44	3.87	5.07	5.61	5.80	6.16	5.05	3.67	2.12	1.13	0.90	(44)	
(45)		RadStd	0.10	0.16	0.27	0.35	0.49	0.42	0.29	0.40	0.32	0.28	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	+1.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page